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Purpose/Policy:

The purpose of this SOP is to assist with SIP and CIP of both MM1 and MM2. It will also help guide you through pulling beer and changing beer process.

Scope:

The keg line Minomats (MM) are responsible for cleaning and filling kegs. The MMs are comprised of 8 stations. The first 7 stations are for cleaning the inside of kegs. The last station is where kegs are filled with beer. Kegs are transferred from one station to another via a pneumatic rail that lifts them up and moves them to the next station.

Definitions/Acronyms:

- MM – Minomats
- SIP – Sanitization in Process
- CIP – Cleaning in Process.
- DAW – (DO Water)

Authorities/Responsibilities:

This SOP can be completed by a lead level packaging operator after proper training from a senior member of the Keg line packaging team. The objective of this SOP is to gain the knowledge needed to operate the MM1 and MM2 SIP and CIP and trouble shoot any issues that may arise.

Resources and Materials:

Operations:

- Kegs (Size varies on different runs)
- Liquid Product
- DAW (DO Water)
- CO2/Nitrogen
- Din Fitting Wrench
- False Kegs
- Plug for Air Line

Changeovers:

- Guide Rails for 5.17gal kegs

Deep Clean:

- Refer to Zone cleaning folder

Troubleshooting:

- No Equipment required.

PPE Required:

- For SIP and CIP Brink – Green Rubber Gloves, Face Shield and Safety Glasses.
- For Foaming – Safety Glasses, Face Shield and Nitrate Gloves
- For CO2 process – Hearing Protection.

Procedures:

SIPing the MMs

Setting up for SIP

1. Start by undoing all E-Stops on the racking line.
2. On either of the HMI screens, go to the main menu.
3. From the main menu press “Production Start” and then press “Peri On”. This will engage the peripherals.
4. Next head down to the brink in the cellars.
5. At the brink we will start cold flushing the line.
6. Make sure the red and grey hoses are in the drain and are open.
7. Ensure that the brink manifold is set up for the racker line and not the can line.
8. Attach the racker product hose to the brink.
9. Attach a cold-water hose to the brink manifold.
10. Open up the rack product line.
11. Open up the brink return valve half-way and wait for flow to start going into the brink.
12. Make sure that the keg line pump is in the off position on the power disconnect.
13. Head back up stairs to the rack line.
14. We will next start setting up to bypass the drain valve.
15. To do this start by opening up the right-hand door of the Minomat to access station 8.
16. Remove the blue plug from the black 90-degree fitting that’s between the “Drain” valve and the “CIP” valve.
17. Disconnect the tubing going into the “Drain” valve and insert it into the 90-degree fitting that you just unplugged.
18. Insert the blue plug into the “Drain” valve.
19. Repeat steps 15 through 18 for both MM1 and MM2.
20. Close both doors to MM1 and MM2.
21. Make sure we are going through the 1 micron filter housing for SIP.



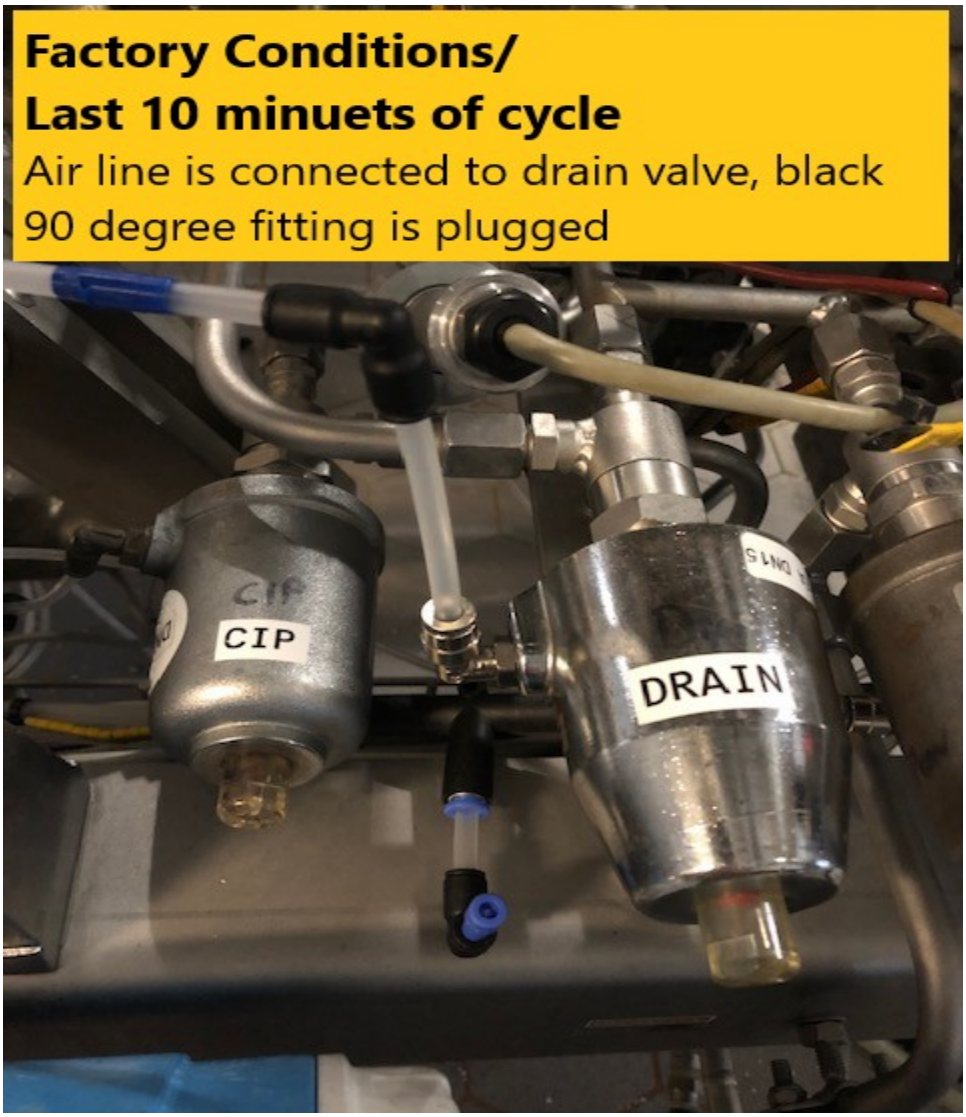
Running SIP

1. On the HMI select “CIP External” in the drop-down menu of the “CIP Setting” screen.
2. Press “CIP On”
3. Confirm that the line has been fully flushed out of any caustic or acid.

4. Close the drain valve for both the drain hose and brink.
5. Let the brink fill with water.
6. Collect 1,500mL (2,000mL if sanitizing hard pipes) of Peroxyacetic Acid (PAA).

Note: Make sure to wear proper PPE when handling chemicals.

7. Once the brink is filled with cold water, close the cold-water hose. Open the valve connecting the brink to the rack product line.
8. Add PAA to the brink.
9. Next turn the keg line product pump disconnect to the “On” position.
10. Press the green button above the switch to start the pump.
11. Now let this cycle run for 40 minutes.
12. While this cycle runs make sure to bleed all ports on the filter housing briefly.
13. If using the 20 micron filters make sure they are isolated from the line and begin hot flushing the 20 micron filter housing. Make sure PAA does not go through the 20 micron filter housing.
14. After the cycle has been going for 30 minutes crack open ball valves behind flow metes on both Minomats. Leave these open for a few minutes. These valves are known as Foam on Beer valves (FOB).
15. Bleed some PAA through the pigtail.
16. After 40 minutes we will set up the Minomats valves for Factory Conditions.
17. Start by stopping the CIP cycle on the HMI.
18. Open the right-hand sliding door of the Minomat to access station 8.
19. Remove the blue plug from the “Drain” valve.
20. Disconnect the tubing from the 90-degree fitting that lies between the “Drain” and CIP valve.
21. Connect the tubing to the “Drain” valve.
22. Insert the pug into the black 90-degree fitting.
23. Steps 18 through 22 must be done for both MM1 and MM2.
24. After both Minomats have been switched over close the doors and turn CIP cycle back on.
25. Run the CIP cycle for another 10 minutes.



Stopping SIP

1. Go to the brink in the cellar and press the red stop button above the keg line pump switch.
2. Open the drain valves for both the brink and the drain hose.
3. Close the valve that connects the brink to the rack product hose.
4. Close the rack product hose valve.
5. Go back to the racker HMI and in the “CIP Setting” screen select “CIP Off”.

DAW (D.O. Water) Rinse:

1. Start in the cellar by flipping the keg line magnet from “CIP Pre-Flux” to “Product”.
2. Take the product line and hook it up to the D.O. drop.
   2.1 If doing hard pipes hook up to either the red or blue line and in the new cellar hook up to the D.O. drop.
3. Check with the filter room operator to make sure you are clear to use D.O. water.
4. If clear to use D.O. water open the D.O. water supply and the keg line product hose.
5. Go to the filter housings on the rack and flush until PAA is all cleared out.
6. On the HMI push the menu button and select “CIP Settings”.
7. Next press “Beer Request”.
8. Lastly press “CIP On”.
9. Now run 200L through both MM1 and MM2.
10. Once both MMs have reached 200L go to the HMI and press “Sterilization End”.
11. Press enter to execute.
12. Return to the cellar and close the D.O. water supply valve and the keg line product valve.

Pulling Beer and Starting:

1. Before starting to pull any liquid over make sure you have the proper sample containers for QC checks.
2. Also make sure you have the correct filter housing in line for the product that is being pulled over.
3. Make sure head pressure is hooked up to the tank you are pulling from.
4. In the cellar open up the tank and the racker product line.
5. On the racker HMI in the CIP setting screen, select “Beer Request” and then press “CIP On”.
6. Watch the conductivity meter for liquid. Once the conductivity mete shows liquid go through all the filter housing ports and bleed them until product comes out.
7. Once you see liquid in the sight glass press “Sterilization End”.
8. From the “Menu” screen select “Flow Settings”.
9. In the “Flow Settings” page, press and hold “Reset” button for 3 seconds. Repeat this process for all “Reset” buttons. This will reset all the numbers in the flow meter window.
10. Press “M2 Flow Meter” and repeat step 9.
11. Return to the “CIP Screen” and then pressing “Sterilization Settings”.
12. Next press “Beer Request – Enter” and “CIP On”.
13. Run off 75L on both Minomats. You can see this from each Minomat’s fill station control panel.
14. Once both Minomats have finished running off their 75L press “CIP Off” on the HMI.

15. The false kegs on station8 will sanitize at this point. Let this step run until the screen reads “Sterilization Finished” on the “CIP Screen”.
16. Then press “Sterilization End”.
17. Remove the false kegs.

Note: Make sure on the “CIP Screen” it reads “Sterilization Finished”. If the false kegs are removed before sterilization is finished there is a risk of being burned by high pressure steam.

18. Press the “Production Preselect” drop down menu.
19. Select the correct size for the kegs that are being run.
20. Press “Production Mode 240 Keg/h”
21. Start MM1 first by pressing “Start” on fill station control panel. Do this for MM2 as well.
22. Once the first keg on each Minomat is filled and pushed out, pull it off the line for qualification.
23. Once the second keg on each Minomat starts to fill press the “Stop” button on the fill station control panel.
24. If both Minomats are qualified press the green “Start” button to start the run.
25. Ensure that both empty and full keg elevators are on.

Finishing a Run and Setting Up for CIP:

1. If the tank has been kicked the rack will throw an alarm signaling this.
    1.1 The faults you should get in the event of kicking a tank are “Maximum Fill Time Exceeded” and “Auto Slow Down by Media Pressure Failure”.
2. In this event press “Failure Screen” in the top left of the HMI.
3. Confirm the faults by pressing “Fault ACK”.
4. Select “Production Start” and press “OFF” on both filling stations 1 and 2.
5. In the cellar on the product pump turn it to the off position.
6. If changing to another beer refer to the beer to beer matrix.
    6.1 Some beers only require a D.O. water flush between them. If this is the case please refer to DAW (D.O. Water) Rinse section of this SOP.
7. If finishing for the day hook up keg line product hose to the brink with a hot water hose also connected to the brink manifold.
8. Open up your water source and your product line.
9. Press the green start button on both Minomat’s fill station control panels.
10. To avoid partial kegs press the red stop button on MM1 fill station control panel.

Note: If pushing from the old cellar: 13.2-gal half barrels, the push will only fill 4 kegs, for logs it will fill 10 kegs.

11. Once the rack has a conductivity fault and the last keg is unable to fill, put the Minomats into passage mode using the HMI.
12. In the “Production Start” menu press “Filling Station 1”.
13. Change the fill station mode to “passage Mode” by pressing on the green “Production Mode” box. This box should change to blue and read “Passage Mode”.
14. Make sure all filled kegs have gone over the scale.
15. Select the “Main Menu” screen on the HMI.
16. Record the full keg count and the scale reject count on paperwork.
17. To clear out any kegs that weren’t filled, go to the HMI and go to the “Main Menu”.
18. Select “Scale Function” tab.
19. Press “Machine Mode”, this will put the scale into passage mode and allow kegs to pass over it.
20. Next press the green start button on the filling station control panel of both Minomats.
21. Once all kegs have been cleared off the line go to the HMI and press “Production Preselect” tab.
22. Using the arrows select “Production Mode Finished” and press enter.
23. Remove the fill heads from station 8 and replace them with the false kegs.
24. Next, we want to flush any remaining beer out of the line.
25. Make sure that “Production Mode” is off in the “Production Preselect” screen.
26. Select “CIP Settings” tab and select "Beer Request”. This will push anything in the lines to the drain in the External Washer.
27. Confirm there is flow in both lines using the flow meters.
28. Once you see color change in the sight glass go to the “Flow Meter” screen on the HMI and record flow meter numbers for the Minomats.

Running CIP:

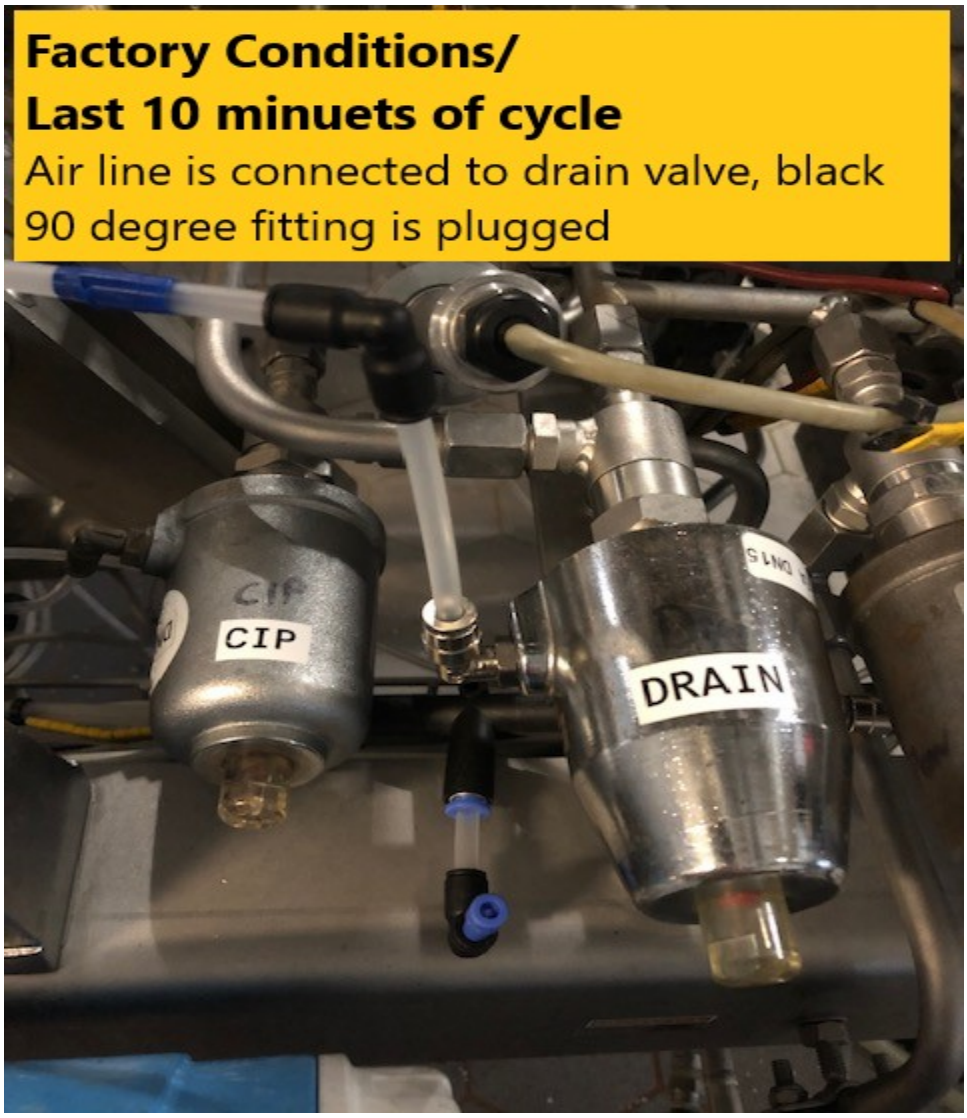
1. Make sure that the brink manifold is set up for the keg line and not the can line.
2. Next make sure the hot hose is hooked up to the drain manifold of the brink as well as the keg line product hose.
3. Make sure both drain hoses for the brink are open and in the cellar drain.
4. Open up the hot hose and the keg line product hose.
5. Go back upstairs to the HMI of keg line.
6. Confirm there is no product in the sight glass on both Minomats.
7. On the HMI select “Sterilization End” and press enter.
8. Now return to the brink and change the magnet from “Production Mode” to “CIP Mode”.
9. Slowly open the brink return valve and make sure that liquid is flowing into the brink.

10. Next you will bypass the drain valves.
    - 10.1 Make sure to stop the CIP cycle.
    - 10.2 Open the right-hand sliding door of the Minomat to access station 8.
    - 10.3 Remove the blue plug from the black 90-degree fitting that sits between the “Drain Valve” and the “CIP Valve”.
    - 10.4 Disconnect the tubing going to the “Drain Valve” and insert it into the 90-degree fitting.
    - 10.5 Insert the blue plug into the “Drain Valve”.
    - 10.6 Follow steps 10.1 through step 10.4 for both Minomats.
    - 10.7



10.8 On the HMI select “CIP External” and press the “Enter” button.
10.9 Press “CIP On”

11. Once the line is clear and only has hot water flowing through it close both drain hoses so that water is returning to the brink. Make sure that the water in the brink stays below 150 degrees and above 120 degrees. Adding Cold water will be necessary to hit this target temperature.
12. On the racker HMI make sure on the “CIP Setting” screen it reads “CIP External in Progress”.
13. While wearing proper PPE collect the amount of caustic for either hard pipes or just the rack line.
    13.1 1.9gal for just the rack line or 2.9gal for hard pipes.
14. Once the brink is full of the correct temperature water close off the water supply valve as well as the valve on the hot hose itself.
15. Open the supply valve on the bottom of the brink and quickly turn the pump disconnect switch to the “On” position and press the green button to start the pump.
16. Confirm that the product pump has started and there is flow into the brink from the return spout.
17. Return to the filter housing and bleed the filter of any air that might be in the head space.
18. Return to the brink and press the red stop button on the product pump. Then quickly close the brink supply valve.
19. Top off the brink with water and add caustic.
20. Now press the green start button on the pump and quickly open the brink supply valve.
21. This is the start of your 40 minute caustic cycle.
22. After 30 minutes crack open the ball valves behind the flow meters on both Minomats. Leave these open for 5 minutes. These valves are know as “Foam On Beer” valves (FOB).
23. Make sure to bleed caustic through the sample port and pig tail.
24. After 40 minutes stop the CIP cycle.
25. Set the Minomat valves to factory condition.
    25.1 Make sure to stop the CIP cycle.
    25.2 Open the right-hand sliding door of the Minomat to access station 8.
    25.3 Remove the blue plug from the “Drain” valve.
    25.4 Disconnect the tubing from the 90-degree fitting that lies between the “Drain” and CIP valve.
    25.5 Connect the tubing to the “Drain” valve.
    25.6 Insert the pug into the black 90-degree fitting.
    25.7 Steps 25.1 through 25.5 must be done for both MM1 and MM2.

25.8 

25.9 After both Minomates have been switched over close the doors and turn CIP cycle back on.

25.10 Run the CIP cycle for another 10 minutes

26. After the last 10 minutes the CIP cycle will be complete.
27. Press the red stop button by the keg line pump disconnect.

Hot Flushing:

1. After completing the CIP the line will need to be hot flushed out.
2. Open up both drains for the brink and close the brink supply valve.

3. Turn on the hot water hose and open up the water supply valve.
4. Let hot water run for at least 20 minutes to make sure the line is clear of chemicals.

Cold Flushing and Acid Cycle:

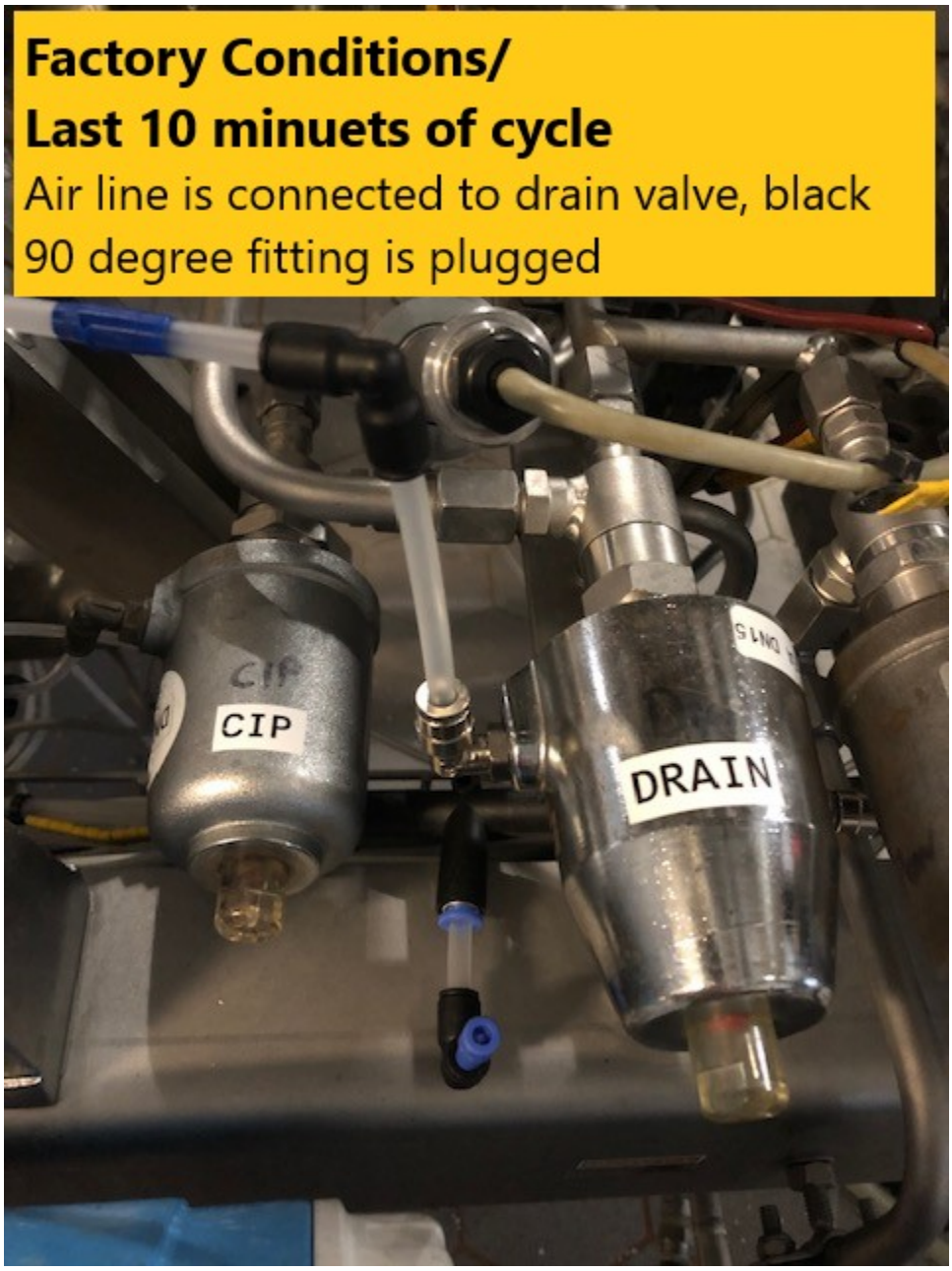
1. After the last run of the week you will need to run an acid cycle after finishing CIP.
2. Turn off and disconnect the hot hose from the brink supply manifold.
3. Hook up a cold water hose to the brink supply manifold.
4. Make sure both drains are open for the brink and the brink supply valve is closed.
5. Open up the valves for your cold water hose and water supply valve. Also make sure your product line is open.
6. Cold flush the line until the water is cold coming out and there is no chemical residue.
7. Go up stairs now to the Minomats.
8. We will set up the Minomats to bypass the drain.
    - 8.1 Make sure to stop the CIP cycle.
    - 8.2 Open the right-hand sliding door of the Minomat to access station 8.
    - 8.3 Remove the blue plug from the black 90-degree fitting that sits between the “Drain Valve” and the “CIP Valve”.
    - 8.4 Disconnect the tubing going to the “Drain Valve” and insert it into the 90-degree fitting.
    - 8.5 Insert the blue plug into the “Drain Valve”.
    - 8.6 Follow steps 10.1 through step 10.4 for both Minomats.

8.7



9. Return down to the brink.
10. Make sure the brink return valve is open and close both drains.
11. While wearing proper PPE collect the amount of Acid for either hard pipes or just the rack line.
    11.1 3gal for just the rack line or 4gal for hard pipes.

12. Once the brink is full of cold water close off the water supply valve as well as the valve on the cold hose itself.
13. Open the supply valve on the bottom of the brink and quickly turn the pump disconnect switch to the “On” position and press the green button to start the pump.
14. Confirm that the product pump has started and there is flow into the brink from the return spout.
15. Return to the filter housing and bleed the filter of any air that might be in the head space.
16. Return to the brink and press the red stop button on the product pump. Then quickly close the brink supply valve.
17. Top off the brink with water and add Acid.
18. Now press the green start button on the pump and quickly open the brink supply valve.
19. This is the start of your 40 minute Acid cycle.
20. After 30 minutes crack open the ball valves behind the flow meters on both Minomats. Leave these open for 5 minutes. These valves are know as “Foam On Beer” valves (FOB).
21. Make sure to bleed Acid through the sample port and pig tail.
22. After the 40 minutes we will set the Minomates back to factory condition.
23. To set the Minomat valves to factory condition follow these steps.
    - 23.1 Make sure to stop the CIP cycle.
    - 23.2 Open the right-hand sliding door of the Minomat to access station 8.
    - 23.3 Remove the blue plug from the “Drain” valve.
    - 23.4 Disconnect the tubing from the 90-degree fitting that lies between the “Drain” and CIP valve.
    - 23.5 Connect the tubing to the “Drain” valve.
    - 23.6 Insert the pug into the black 90-degree fitting.
24. Steps 25.1 through 25.5 must be done for both MM1 and MM2.

25. 

26. Turn the CIP cycle back on and let run for 10 minutes.

Ending CIP/Acid Cycle:

1. On the pump disconnect press the red stop button.
2. Then turn the disconnect to the off position.

3. Close the brink supply valve and the product line valve.
4. Open up both drain valves.
5. Let all water and chemical drain out.
6. Return to the rack HMI and press “Sterilization End” in the “CIP Settings” screen.
7. This will finish allowing the water to drain out.

Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

Not Applicable

Document History and Approval:

Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

VERSION NO.	VERSION DATE	DESCRIPTION OF CHANGE	AUTHOR NAME/PROJECT MANAGER
001	2024/08/02		Kyle Szymanski
002	2026/10/26	Updated SOP to Newest Layout Version	Kyle Szymanski

Version Approval Number: 002

Approved by: Kyle Szymanski

Position Title: OSCI Manager

Date: 10/26/26